

The Performance Commons — List of Figures and Tables

Read top to bottom, the visuals carry the paper's argument on their own: magnitude → mechanism → the same problem, distributed → generality → proof → destination → already underway. Each entry notes the beat it carries and whether it rests on measured data or is an argued/conceptual construct.

Table 1 — The efficiency gap, quantified. (§1) The measured cost of generality: ASIC 1×, CGRA ~2.8× area, FPGA ~20–35× area / ~10–14× power, CPU ~100–500× energy. *Beat: the gap to ASIC-level performance is real and large.* — **quantitative, sourced** (Hameed; Kuon & Rose; Prabhakar).

Figure 1 — The general-purpose vs. custom trade-off is mostly an effort artifact. (§2) The effort gap, concentrated on reconfigurable substrates; close it and a CGRA sits within ~2.8× of an ASIC with full flexibility. *Beat: that gap is effort, not physics — the hardware question (Q1).* — **anchored endpoints**; finite-effort curve illustrative.

Figure 2 — Distributed computing is a collective-action problem, not a hardware limit. (§3) The coordination gap by looseness of coupling; naive methods waste almost all of a loose federation's compute, while methods built for loose coupling (DiLoCo: ~500× less communication, ~90% utilization across continents) reclaim it. *Beat: the same failure one level up — the wasted-supply side of the software question (Q2).* — **anchored to DiLoCo / OpenDiLoCo**; the underlying waste (12–18% average server utilization, up to ~30% "comatose") is sourced (NRDC / Anthesis).

Figure 3 — Machine demand has no floor. (§3) The value of the marginal compute-hour against cumulative demand: human demand saturates, machine demand extends toward arbitrarily low value — past datacenter economics (~\$0.50/hr) into the range only cheap idle compute (~\$0.08/hr, ~6× cheaper) can serve. *Beat: why solving it is worth the effort — unbounded machine demand makes the otherwise-wasted resource valuable.* — **conceptual**; the cost anchors are from Omerta's reliability-market simulation, the task-value ladder from its economic analysis.

Table 2 — One pattern, every layer of the stack. (§6) Foundry → CAD/EDA → IP → ISA → firmware → kernels → drivers → measurement data → distributed compute, each with its open counter-trend. *Beat: it is not one gap — the same coordination failure recurs from the foundry to the ISA.* — **qualitative + quantitative.**

Table 3 — Open commons with higher relative evolutionary fitness. (§7) Linux, LLVM, RISC-V, Mesa, PyTorch, FFmpeg — what each displaced and the evidence of dominance. *Beat: we have solved exactly this before — the commons demonstrably wins, at six different layers.* — **quantitative dominance.**

Figure 4 — Where the optimum lands as the commons matures. (§10) A phase diagram over commons-maturity-over-time × workload-diversity; the reconfigurable-optimal region sweeps into territory that today defaults to general-purpose silicon. *Beat: over time the optimum for real workloads migrates into reconfigurable silicon — not fixed ASICs.* — **conceptual**; the boundary is exactly what a bigger whole-system characterization (§11) would pin down.

Figure 5 — Reconfigurable hardware is already moving in; the commons is the inflection. (§10) A decade of reconfigurable entry into the datacenter — Catapult, Intel–Altera \$16.7B, AWS F1, AMD–Xilinx \$49B, the

AI-accelerator wave — with a projected inflection as the programmability gap closes. *Beat: the migration is measurable and already expensive; machine intelligence is the accelerant.* — **real milestones + acquisition values**; share curve illustrative, post-2025 a projection.

How the sequence maps to the prose

Visual	Section	Argument beat
Table 1	§1	the gap is real and large
Figure 1	§2	the gap is effort, not physics (Q1: hardware)
Figure 2	§3	distributed computing is the same problem — the wasted supply (Q2: software)
Figure 3	§3	unbounded machine demand makes the idle resource valuable
Table 2	§6	the same failure is everywhere in the stack
Table 3	§7	the commons has solved it before, and wins
—	§8–§9	(AI multiplies the commons; the vendor is the aggregation point — prose only)
Figure 4	§10	where the optimum lands as the commons matures
Figure 5	§10	and it is already underway
—	§11 / Conclusion	honest limits; build collaboration-native hardware (prose only)

The "needs a bigger project" items (flagged for honesty)

- **Figure 4's boundary** — the real position of the ASIC / CPU / CGRA optimal regions needs the whole-system workload characterization §11 calls the prerequisite (achieved-vs-achievable parallelism, dataflow locality, phase and input statistics across the real consumer mix).
- **Figure 3's demand curve** — conceptual; the ~\$0.50 vs ~\$0.08 cost anchors are from Omerta's reliability-market simulation and the task-value ladder from its economic-analysis notes. Omerta itself is partially built: a substantial, well-tested mesh-networking layer (encryption, NAT traversal, gossip discovery, tunnelling, plus Terraform-managed bootstrap/STUN servers) and working ephemeral-VM and simulation components, but the end-to-end compute loop is mid-migration and the trust/payment chain is validated in simulation rather than running live (two of six transaction types tested).
- **Figure 5's share curve and projection** — the adoption trend is illustrative; the dated milestones and the \$16.7B / \$49B acquisition values are real.
- **Figure 1's finite-effort curve** — illustrative; only the realizable-frontier endpoints (the intrinsic fabric tax) are measured.